

SMART INDIA HACKATHON 2026 // PS: SIH26066

SIH26066: OCEANEMBED

Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations

Comprehensive Team Master Decision & Execution Blueprint (2026 Edition)

Document Control & Classification:

- **Problem Statement ID:** SIH26066 (Ministry of Earth Sciences / INCOIS Domain)
- **Geographic Domain:** North Indian Ocean (5°N - 30°N, 45°E - 105°E)
- **Spatial Grid:** 0.25° × 0.25° Resolution (101 × 241 Grid, 24,341 Spatial Cells)
- **Vertical Depths:** 15 Standard Levels (0.494m to 1000m)
- **Document Version:** v3.0 (Strictly Audited & Empirically Verified)
- **Target Audience:** All 6 SIH Team Members, Faculty Mentors, and Evaluation Committee
- **Classification:** CONFIDENTIAL // INTERNAL TEAM MASTER BLUEPRINT
- **Date of Empirical Audit:** September 2026
- **Lead Contributors:** SIH26066 Core Team (Data Engineering, Ocean Geophysics, Deep Learning, Scientific Validation, Systems Integration)

EXECUTIVE DIRECTIVE FOR ALL TEAM MEMBERS

This document is not an aspirational pitch or a generic literature summary. It is the definitive, empirically audited engineering blueprint for problem statement **SIH26066**. Every claim regarding dataset availability, tensor dimensions, vertical interpolation, land masking, and computational overhead in this guide has been verified against live NetCDF data and executing Python/PyTorch code in our local workspace.

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Core Principle: Absolute scientific and empirical honesty. We do not manufacture convergence numbers, fake benchmark baselines, or unverified satellite coverage. All components are labeled with their strict audit status: 'VERIFIED', 'PILOT-VERIFIED', 'OFFICIAL-CATALOG-VERIFIED', 'PLANNED', or 'OPEN RISK / UNVERIFIED'.

PART 2: EXECUTIVE DECISION SUMMARY

SIH26066 FINAL TEAM DECISION MATRIX

RECOMMENDATION: GO — LOCK SIH26066 FOR SMART INDIA HACKATHON
DECISION STATUS: UNANIMOUS FINAL APPROVAL BASED ON EMPIRICAL FEASIBILITY AUDIT
TECHNICAL PATH: DEMONSTRATED & CODE-VERIFIED (NO FATAL BLOCKERS REMAIN)
SCIENTIFIC RISK: DEEP SUBSURFACE IDENTIFIABILITY (FULLY MANAGEABLE WITH CLEAR FALLBACKS)

1. Executive Summary

The technical debate regarding whether our team should commit to **SIH26066** (*Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations*) is now officially closed. Following an exhaustive 12-phase data feasibility and code-level audit, our team has established that the data ingestion pipeline, spatial regridding, vertical interpolation, tensor assembly, masked loss formulation, and independent ARGO ground-truth validation are **100% operational**.

2. Key Audit Milestones Established

- Target 3D Reanalysis Verified:** We successfully downloaded and verified the official 3D GLORYS reanalysis pilot (`cmems_mod_glo_phy_my_0.083deg_P1D-m`, 44.74 MB). It spans 36 native depth levels down to 1062.44m, which we vertically interpolated to the exact 15 target depths specified by SIH (0.49m to 1000m) and regridded to the 0.25° grid (101×241) in 1.92 seconds.
- Real 14-Channel Surface Input Constructed:** All seven physical surface variables (SST, SSS, SSH, Current U, Current V, Wind U, Wind V) plus seven corresponding binary validity masks have been sourced from genuine satellite/multi-obs L4 products and assembled into `data/pilot/sample_X_Y_real.pt` (2.82 MB).
- Genuine Satellite SSS Confirmed:** Sea Surface Salinity is sourced from the official satellite Multi-Observation L4 product (`cmems_obs-mob_glo_phy-sal_my_multi-oi_P7D-c`, 488 KB pilot). It was regridded to 101×241 and weekly-to-daily interpolated. Channel 1 of our real input tensor contains genuine physical values (27.86 to 44.13 psu, mean 34.57 psu) with zero NaNs.
- Independent ARGO Pipeline Functional:** We downloaded an official multi-profile ARGO NetCDF (`20221101_prof.nc`, 6.14 MB) from the Coriolis Global Data Assembly Centre (GDAC). We isolated Indian NIO floats (Float #34 at 17.9540°N , 63.5540°E), colocated it to our grid cell (18.00°N , 63.50°E), and interpolated its real CTD measurements across all 15 depths down to 1000m.
- Land Mask Loss Isolation Mathematically Proven:** The North Indian Ocean domain contains 12,487 land cells (51.30%) and 11,854 valid ocean cells (48.70%). We proved via live script injection that injecting extreme corrupted values (999,999.0) into land cells yields exactly 0.000 change in loss and gradients.
- Integration Sanity Confirmed:** Real-data training-step integration passed across Pointwise MLP, Simple CNN, and OceanEmbedNet architectures. All 7 adversarial stress tests in `scripts/stress_test_break_everything.py` passed with zero errors.

3. Immediate Action Mandate

The team will not waste time reopening debate on alternative problem statements. We lock SIH26066, assign the six operational roles detailed in this document, and execute Phase 1 of historical corpus downloading.

PART 3: THE PROBLEM STATEMENT IN PLAIN ENGLISH

1. The Core Oceanographic Challenge

Imagine standing on the deck of a research vessel in the Arabian Sea or Bay of Bengal. The water surface looks uniform, yet beneath the surface lies a complex, layered thermodynamic engine that dictates global weather, monsoons, and cyclone formation.

The ocean water column is divided into three primary zones:

- **The Epipelagic / Mixed Layer (0 to ~50–100m):** Driven by surface atmospheric winds, solar radiation, and evaporative cooling. Here, temperature is relatively uniform due to turbulent mechanical mixing.
- **The Thermocline (100m to ~300–500m):** The critical transition layer where water temperature plunges dramatically from ~28°C to below 10°C across just a few hundred meters. The thermocline acts as a dynamic thermal barrier and energy reservoir.
- **The Deep Abyssal Ocean (500m to 1000m+):** A cold, dense, slowly circulating water mass with temperatures gradually tapering from 8°C down to 4°C or lower.

2. The Satellite Blindspot

Earth observation satellites in low-Earth and geostationary orbits orbit hundreds of kilometers above the ocean. They carry advanced radiometric, radar, and optical sensors. However, **electromagnetic radiation cannot penetrate seawater**:

- Infrared radiometers (measuring SST) only penetrate the upper skin layer of the ocean (10 to 20 micrometers).
- Microwave radiometers (measuring SST and SSS) penetrate the sub-skin layer to roughly 1 millimeter or 1 centimeter.
- Radar altimeters (measuring SSH) measure the physical height of the sea surface relative to the reference geoid.
- Scatterometers (measuring wind stress) reflect off capillary and gravity surface waves (1 to 5 cm ripples).

The central paradox: Satellites provide comprehensive, daily, basin-wide horizontal observations of the surface skin, but are physically blind to the subsurface ocean. Conversely, physical research vessels and in-situ instruments provide vertical depth profiles but are sparse, expensive, and leave massive observational voids across time and space.

3. How AI and Physics Enable Subsurface Inference

If satellites cannot see underwater, how can any AI model reconstruct temperatures down to 1000 meters?

The answer lies in **ocean geophysical coupling**:

1. **Geostrophic Balance & Dynamic Topography:** A deep pool of warm water expands thermal volume, creating a localized dome on the ocean surface measured by satellite altimetry (Sea Surface Height Anomaly, SLA). Conversely, deep cold upwelling causes surface depressions.
2. **Baroclinic Modes & Internal Waves:** Internal density variations propagate as baroclinic Rossby and Kelvin waves, which modulate both surface currents (u, v) and sea level.
3. **Ekman Pumping & Wind Stress Curl:** Atmospheric wind vectors drive horizontal surface divergence or convergence (Ekman transport), forcing vertical suction (upwelling of cold thermocline waters) or downwelling (pumping warm surface waters into the depths).
4. **Salinity Stratification & Steric Height:** Sea Surface Salinity (SSS) combined with SST determines surface water density. In the Bay of Bengal, massive river runoff (Ganges-Brahmaputra) creates fresh surface layers that decouple surface heat from the thermocline, creating barrier layers.

OceanEmbed bridges this gap: By ingesting 7 multi-modal surface observational fields over spatial neighborhoods, the model extracts latent physical footprints of deep dynamical processes and projects them vertically to reconstruct the complete 0-1000m temperature profile.

PART 4: WHY THIS PROBLEM MATTERS (SCIENTIFIC & STRATEGIC VALUE)

1. Cyclone Rapid Intensification (RI) in the North Indian Ocean

The Bay of Bengal and Arabian Sea host some of the most destructive tropical cyclones on Earth (e.g., Cyclones Amphan, Tauktae, Fani, Biparjoy). Operational weather forecasts frequently fail to predict **Rapid Intensification (RI)**—when a cyclone's wind speed increases by ≥ 30 knots in 24 hours.

- Surface SST alone is insufficient to predict cyclone intensity because cyclone-induced winds stir up water from 50-100m depth.
- If the subsurface thermocline is shallow and cold, the cyclone quickly upwells cold water, cooling the sea surface and choking its own energy supply (negative feedback).
- If the subsurface harbors a deep warm pool with high **Ocean Thermal Energy (OTE)** or **Tropical Cyclone Heat Potential (TCHP)**, wind stirring only brings more 28°C water to the surface, supercharging the storm into a Category 4 or 5 monster.
- **OceanEmbed enables basin-wide, daily maps of TCHP and thermocline depth, giving disaster management agencies critical 48-hour advance warnings of rapid intensification.**

2. Monsoon Dynamics & Indian Ocean Dipole (IOD)

The Indian Summer Monsoon Rainfall (ISMR) directly sustains 1.4 billion people and Indian agriculture. The monsoon is coupled with the **Indian Ocean Dipole (IOD)** and the **Madden-Julian Oscillation (MJO)**, both of which are governed by east-west subsurface heat content shifts across the equatorial Indian Ocean. Rapid 3D temperature reconstruction provides direct data assimilation inputs for climate forecast models.

3. Naval Anti-Submarine Warfare (ASW) & Acoustic Duct Modeling

Seawater temperature directly governs the speed of sound via the Mackenzie equation:

$$c(T, S, z) = 1448.96 + 4.591 T - 5.304 \times 10^{-2} T^2 + 2.374 \times 10^{-4} T^3 + 1.340 (S - 35) + 1.630 \times 10^{-2} z + \dots$$

- Sharp temperature gradients in the thermocline create **Sound Velocity Profiles (SVPs)** that refract sonar waves, forming **acoustic shadow zones** where submarines can hide undetected, or **sound channels (SOFAR channels)** that channel acoustic signals over thousands of kilometers.
- The Indian Navy currently relies on sparse bathythermograph (XBT) drops or coarse climatology. A daily 0.25° 3D temperature field gives naval tacticians real-time acoustic propagation maps.

4. Marine Fisheries & Primary Productivity

Marine fisheries in the Arabian Sea (e.g., along the Malabar coast and Oman upwelling zone) depend on nutrient-rich upwelling from below the thermocline. Mapping the upward displacement of the 20°C isotherm (D_{20}) pinpoints potential fishing zones (PFZs), providing tangible economic benefits to coastal communities.

PART 5: WHY WE SELECTED SIH26066 (COMPETITIVE LANDSCAPE ANALYSIS)

1. Landscape Comparison: SIH26066 vs. Typical SIH Problem Statements

Every year at the Smart India Hackathon, hundreds of teams select problem statements from categories such as:

- Generic AI/ML Chatbots (e.g., student grievance portals, healthcare FAQs)
- Web/Mobile Management Portals (e.g., hostel room allocation, inventory trackers)
- Standard Computer Vision (e.g., traffic violation detection, pothole detection on public roads)

Assessment Dimension	Generic SIH Problem Statements	SIH26066 (OceanEmbed)	Strategic Advantage for Our Team
Scientific Depth	Low: Generic CRUD or standard YOLO/ResNet wrapper	Very High: Satellite geophysics, fluid dynamics, multi-modal sensor fusion	Immediate differentiator in front of senior academic and ministry juries
Data Authenticity	Often simulated, scraped, or unverified CSV files	Real Copernicus Marine & Coriolis GDAC NetCDF satellite and in-situ data	High credibility; jury cannot dismiss as a 'toy mock' project
Evaluation Objectivity	Subjective (juries judge UI aesthetics, feature lists)	Strictly Quantitative: Depth-by-depth RMSE (^°C), bias, ARGO colocation	Hard numerical proof of model superiority over baselines
Competitive Density	Extreme: 40–80 teams submit nearly identical apps	Moderate to Low: High barrier to entry filters out low-effort teams	Juries remember technically rigorous projects that address national priorities
Ministry Alignment	Often disconnected from primary agency operations	Directly aligned with Ministry of Earth Sciences (MoES) & INCOIS mandate	Direct potential for national adoption and post-hackathon incubation

2. Why Our Team Has an Unfair Advantage

1. **We Have Solved the Ingestion Barrier:** Other teams will spend the first 3 days of the hackathon struggling to authenticate Copernicus Marine API, parsing NetCDF-4 files, or discovering that their target files lack 3D depth. We have already downloaded, verified, and assembled the 14-channel input and 15-channel target tensors.
2. **Defensible Scientific Stance:** Most student teams claim '99% accuracy' on synthetic data. We arrive with an audited codebase, mathematically proven land-mask isolation, and an independent ARGO validation pipeline, demonstrating genuine engineering maturity.
3. **National Strategic Resonance:** Reconstructing subsurface ocean temperatures in the North Indian Ocean directly serves India's Deep Ocean Mission, Blue Economy initiatives, and disaster resilience frameworks.

PART 6: WHY THIS WAS NOT A RANDOM CHOICE (DECISION FRAMEWORK)

Our team utilized a multi-criteria weighted scoring framework to evaluate competing SIH problem statements before committing to SIH26066.

1. Decision Evaluation Matrix

Decision Criterion	Weight	Score (1-10)	Weighted Score	Justification & SIH26066 Assessment
Scientific & Technical Depth	20%	9.5	1.90	Combines satellite remote sensing, geophysical fluid dynamics, and multi-scale deep learning.
Data Accessibility & Legality	20%	9.0	1.80	Free, public, authorized operational APIs (Copernicus Marine, Coriolis GDAC) under open scientific licenses.
Feasibility of Demonstration	15%	8.5	1.28	Complete 3D volumetric fields can be rendered interactively via WebGL/deck.gl with slice-based inspection.
Defensibility in Judging	20%	9.5	1.90	Quantitative depth-by-depth RMSE against independent ARGO floats eliminates subjective judging biases.
National Strategic Impact	15%	9.0	1.35	Supports INCOIS, Indian Navy ASW, MoES Cyclone Early Warning, and fisheries management.
Engineering De-risking Already Done	10%	9.5	0.95	End-to-end pilot pipeline, real 14-ch tensor, regridding, vertical interpolation, and training step verified.
TOTAL WEIGHTED SCORE	100%	—	9.18 / 10.0	HIGHEST ACROSS ALL EVALUATED PS CANDIDATES

2. Comparison with Alternative PS Candidates

- *Candidate B (Healthcare Symptom Checker):* Disqualified due to lack of verifiable clinical ground truth, massive medical liability, and extreme competitive saturation (over 60 teams submitting standard LLM prompts).
- *Candidate C (Smart Traffic Signal Optimization):* Disqualified due to synthetic traffic simulation data (SUMO), lack of live municipal sensor feeds, and low technical defensibility.
- *SIH26066 Outcome:* Unambiguous winner based on verified data access, strategic relevance, and deep engineering substance.

PART 7: HOW DIFFICULT IS THIS? (BRUTALLY HONEST DIFFICULTY PROFILE)

We do not underestimate this problem statement. Reconstructing the interior of a turbulent, stratified fluid from surface measurements is mathematically ill-posed.

SIH26066 DIFFICULTY BREAKDOWN		
Domain Component	Difficulty	Primary Engineering & Scientific Bottlenecks
Data Engineering	HIGH	Multi-sensor spatial/temporal harmonization, regridding
Oceanographic Physics	HIGH	Non-monotonic stratification, barrier layers, inversions
Deep Learning Design	MED-HIGH	Multi-scale masked spatial encoding, depth conditioning
Scientific Validation	HIGH	4D spatial-temporal ARGO colocation, strict depth metrics
UI & System Integration	MEDIUM	Fast 3D volume slicing, real-time inference serving

1. Specific Technical Complexities

- The Inversion Problem (Ill-Posed Mapping):** The mapping from 2D surface fields to 3D subsurface profiles is a Fredholm integral equation of the first kind—multiple subsurface thermal configurations can produce nearly identical surface signatures. The model must learn spatial context (gradients, vorticity, divergence) across hundreds of kilometers to infer vertical displacement.
- Coordinate & Grid Harmonization:** Ingesting 7 disparate products with native resolutions ranging from 0.05° (OSTIA SST) to 0.25° (Currents) requires robust conservative/bilinear regridding to the target 101 × 241 grid without introducing interpolation artifacts.
- Missing Data & Swath Gaps:** Scatterometer and altimeter products contain orbital gaps. The model cannot simply fill missing values with zero without introducing artificial thermal shocks. Explicit validity masking is required.
- Computational Scale:** A 15-year daily corpus over 101 × 241 with 14 input channels and 15 target depth channels represents ~ 5,475 daily snapshots (~ 15 GB of packed float32 tensors).

PART 8: WHAT WE HAVE ALREADY PROVEN (AUDITED EVIDENCE)

Every milestone listed in this table has been executed, timed, and logged on real data in our workspace.

Component / Claim	Mechanism & Test Script	Audited Findings & Measured Values	Audit Status
SIH Requirements	Official Problem Statement	Domain: 5°N - 30°N , 45°E - 105°E ; 0.25° grid (101×241); 15 depths.	VERIFIED
3D GLORYS Pilot	<code>scripts/download_glorys_3d_pilot.py</code>	Downloaded 44.74 MB NetCDF (<code>data/pilot/glorys_3d_pilot.nc</code>). 36 native depth levels (0.494m to 1062.44m).	PILOT-VERIFIED
15-Depth Target Pipeline	<code>scripts/process_real_3d_glorys.py</code>	Vertically interpolated to 15 standard depths and regridded to 101×241 in 1.92s. Ocean cells: 11,854 (48.70%).	VERIFIED
0.25° Target Grid	<code>scripts/verify_grid_regridding.py</code>	GLORYS regridded in 0.745s, OSTIA regridded in 0.739s. Verified dimensions: 101×241 .	VERIFIED
SST Dataset	<code>scripts/verify_sst_regridding.py</code>	Ingested real 414 MB OSTIA file (<code>METOFFICE-GLO-SST-L4-REP-OBS-SST</code>). Regridded to 101×241 .	PILOT-VERIFIED
Genuine Satellite SSS	<code>scripts/rebuild_real_xy_with_genuine_sss.py</code>	Downloaded 488 KB pilot (<code>cmems_obs-mob_glo_phy-sal_my_multi-oi_P7D-c</code>). Weekly-to-daily interpolated into Channel 1 (27.86 to 44.13 psu).	PILOT-VERIFIED
SSH Dataset	<code>scripts/download_surface_pilots.py</code>	Downloaded 796 KB DUACS SLA NetCDF (<code>cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.125deg_P1D</code>).	PILOT-VERIFIED
Surface Currents (U, V)	<code>scripts/download_surface_pilots.py</code>	Downloaded 411 KB Multi-Obs surface currents (<code>cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m</code>).	PILOT-VERIFIED
Surface Winds (U, V)	<code>scripts/download_winds_pilot.py</code>	Downloaded 9.24 MB hourly scatterometer winds (<code>cmems_obs-wind_glo_phy_my_14_0.125deg_PT1H</code>). Daily averaged.	PILOT-VERIFIED
Real 14-Ch X & 15-Ch Y	<code>scripts/rebuild_real_xy_with_genuine_sss.py</code>	Assembled 100% genuine observations into <code>data/pilot/sample_X_Y_real.pt</code> (2.82 MB). Verified shapes [1, 14, 101, 241] and [1, 15, 101, 241].	VERIFIED
Land Mask Loss Isolation	<code>scripts/rebuild_real_xy_with_genuine_sss.py</code>	Injected 999,999.0 into 12,487 land pixels. Proved land gradient contribution is exactly 0.000000.	VERIFIED
Training-Step Sanity	<code>scripts/rebuild_real_xy_with_genuine_sss.py</code>	Forward-backward pass executed across Pointwise MLP, Simple CNN, and OceanEmbedNet. Loss: 509.06.	VERIFIED
Independent ARGO Colocation	<code>scripts/audit_argo_real.py</code>	Downloaded 6.14 MB ARGO NetCDF (<code>20221101_prof.nc</code>) from Coriolis GDAC. Filtered Float #34, colocated, and interpolated to 15 depths.	PILOT-VERIFIED
Measured Compute Benchmark	<code>scripts/benchmark_compute.py</code>	Measured on local CPU machine (B=4): Step time 5,133.83 ms, Peak RAM 1.66 GB.	MEASURED
Adversarial Stress Tests	<code>scripts/stress_test_break_everything.py</code>	7 stress tests executed. Fixed IEEE 754 NaN loss bug and batch scaling bug. All 7 tests passing.	VERIFIED

PART 9: WHAT WE HAVE NOT PROVEN YET (SCIENTIFIC BOUNDARIES)

In adherence to strict scientific honesty, the team explicitly defines what has **NOT** been proven yet. No team member or presentation slide may claim these as completed facts:

CRITICAL SCIENTIFIC BOUNDARIES & LIMITS

1. FINAL MODEL CONVERGENCE IS NOT PROVEN:
We executed a single training step on a pilot tensor to verify pipeline data flow. This proves code correctness, NOT that the model has learned ocean physics or reached optimal weights.
2. BEATING BASELINES IS NOT YET PROVEN:
OceanEmbed has not yet been benchmarked against Monthly Climatology, Linear Ridge, or U-Net on a multi-year held-out test set. Superiority is a hypothesis to be experimentally validated.
3. ACCURATE 1000m RECONSTRUCTION IS NOT PROVEN:
Outputting a 15-depth tensor with 1000m dimensions is technically functional. However, physical accuracy at 1000m remains an open scientific question due to surface decorrelation.
4. 15.6 YEARS OF HISTORICAL TENSORS DO NOT YET EXIST ON DISK:
The 2010-2025 overlap is verified in the Copernicus catalog. Downloading and pre-processing the full multi-year training corpus is an upcoming Phase 1 execution task.

Every claim we present to judges will strictly maintain the boundary between **verified engineering infrastructure** and **in-progress scientific training**.

PART 10: OUR SOLUTION — THE OCEANEMBED ARCHITECTURE

OceanEmbed is an end-to-end deep learning framework designed specifically to respect oceanographic physics and multi-modal satellite data properties.

1. Conceptual Architecture Diagram



2. Core Functional Modules

- Masked Multi-Scale Encoder:** Ingests the 14-channel input. Uses gated or partial convolutions where invalid/land pixels do not bleed spurious edge signals into ocean cells. Extracts features across multiple spatial scales (3×3 , 5×5 , 7×7 receptive fields) to capture local frontal gradients and meso-scale eddies (50-200 km).
- Latent Ocean Embedding (Z):** Compresses the 14 surface channels into a structured 128-dimensional latent representation per ocean grid cell. This embedding encapsulates the local thermodynamic and kinematic state (divergence, vorticity, steric height anomaly).
- Depth-Conditioned Decoder:** Rather than treating 15 depth levels as 15 independent, disconnected outputs, the decoder conditions on continuous or learned depth queries. It enforces vertical continuity, modeling the transition from the mixed layer through the thermocline into the abyssal ocean.

PART 11: WHY WE USE MASKS (LAND & MISSING OBSERVATION BOUNDARIES)

In terrestrial computer vision, every pixel in an image is usually valid. In satellite oceanography, this assumption fails catastrophically.

1. The Reality of Ocean Data Gaps

- The Land Mask Barrier:** In the North Indian Ocean domain (5°N - 30°N , 45°E - 105°E), the Indian subcontinent, Arabian Peninsula, Southeast Asia, and islands occupy **12,487 grid cells**, or **51.30% of the entire grid**. Only **11,854 grid cells (48.70%)** are actual ocean water.
 - If land pixels are filled with zeros or arbitrary values, standard convolutional filters will calculate massive spurious gradients at coastlines.
 - The model will spend its parameter budget memorizing coastal outlines rather than ocean physics.
- Cloud Contamination in Optical/Infrared SST:** Infrared satellite radiometers cannot penetrate clouds. During the summer monsoon (June–September), cloud cover in the Bay of Bengal can exceed 80%, creating missing observation swaths.
- Orbital Altimeter & Scatterometer Swath Gaps:** Radar altimeters only measure along narrow nadir ground tracks. Multi-satellite L4 products interpolate these gaps, but coastal and high-latitude coverage varies.

2. The Dual-Channel Solution: Value + Validity Mask

For each of the 7 physical surface variables, we provide two coupled channels:

Input Channel i = Physical Value (normalized)

Input Channel $i+7$ = Binary Validity Mask (1.0 = Valid Ocean Observation, 0.0 = Missing or Land)

```
Channel 0: SST Value      ---> Channel 7: SST Mask
Channel 1: SSS Value      ---> Channel 8: SSS Mask
Channel 2: SSH Value      ---> Channel 9: SSH Mask
Channel 3: Current U Value ---> Channel 10: Current U Mask
Channel 4: Current V Value ---> Channel 11: Current V Mask
Channel 5: Wind U Value   ---> Channel 12: Wind U Mask
Channel 6: Wind V Value   ---> Channel 13: Wind V Mask
```

3. Mathematical Formulation of Masked Loss

Let $Y_{\{d, i, j\}}$ be the ground truth temperature at depth d and coordinate (i, j) , $\hat{Y}_{\{d, i, j\}}$ be the model prediction, and $M_{\{i, j\}} \in \{0, 1\}$ be the ocean validity mask ($M_{\{i, j\}} = 1$ if ocean, 0 if land). The masked mean squared error loss is defined as:

$$L_{\{ocean\}} = \frac{1}{\sum_{d=1}^{15} \sum_{i=1}^H \sum_{j=1}^W M_{\{i, j\}}} \sum_{d=1}^{15} \sum_{i=1}^H \sum_{j=1}^W M_{\{i, j\}} \cdot \left(Y_{\{d, i, j\}} - \hat{Y}_{\{d, i, j\}} \right)^2$$

Mathematical Proof of Land Isolation:

$$\frac{\partial L_{\{ocean\}}}{\partial \hat{Y}_{\{d, i_{\{land\}}, j_{\{land\}}\}}} = 0$$

Because $M_{\{i_{\{land\}}, j_{\{land\}}\}} = 0$, any prediction or corruption over land yields exactly zero loss and zero gradient update.

PART 12: WHY OCEAN EMBEDDING? (THE LATENT SPACE CONCEPT)

1. Demystifying the "Ocean Embedding"

We must be clear with our team and the SIH jury:

****The Latent Ocean Embedding is NOT a metaphysical or "true hidden ocean state."****

Rather, it is a ****learned, compact, multi-scale feature representation**** of the surface dynamical footprint that provides the optimal conditioning context for vertical profile reconstruction.

2. The Information Bottleneck

A direct pointwise mapping from 7 surface values ($T_{\{sfc\}}, S_{\{sfc\}}, \eta, u, v, w_u, w_v$) to 15 subsurface depths lacks spatial context. For instance, a sea surface height anomaly $\eta = +15$ cm could indicate:

- A warm-core anti-cyclonic eddy with a deep thermocline bowl.
- A coastally trapped Kelvin wave propagating along the boundary.
- A broad seasonal steric warming event.

A pointwise model cannot distinguish between these three physical regimes because it sees only a single pixel.

The **Ocean Embedding Network (E_{θ})** looks at an extensive spatial receptive field (150-300 km across the 101×241 grid) using multi-scale convolutional kernels. It computes spatial derivatives:

- Relative vorticity: $\zeta = \partial v / \partial x - \partial u / \partial y$
- Horizontal divergence: $\delta = \partial u / \partial x + \partial v / \partial y$
- Thermal and haline frontal gradients: $|\nabla SST|, |\nabla SSS|$

The encoder compresses these multi-scale spatial gradients into a 128-dimensional embedding vector $Z_{\{i, j\}} \in \mathbb{R}^{128}$ at each grid cell. This embedding represents the local dynamical regime (e.g., eddy center, boundary current, upwelling zone) before vertical reconstruction begins.

PART 13: WHY DEPTH-CONDITIONED DECODING?

1. The Pitfall of Disjoint Multi-Head Regressors

A naive neural network approach to 3D ocean reconstruction uses a standard 2D CNN that directly outputs 15 output channels:

$$\hat{Y} = \text{CNN}(X) \in \mathbb{R}^{15 \times 101 \times 241}$$

Why is this suboptimal?

1. **Treats Depths as Disjoint Classes:** The final 1×1 convolutional layer simply projects features into 15 static channels. The network has no intrinsic mathematical concept that channel 3 (20m) is physically adjacent to channel 4 (30m).
2. **Inability to Generalize to Arbitrary Depths:** If a user or naval operator requests temperature at 125m (between standard levels 100m and 150m), a static 15-channel CNN cannot evaluate it without external interpolation.
3. **Gradient Decoupling:** Errors in the thermocline do not backpropagate smoothly into the mixed layer representation.

2. The Depth-Conditioned Continuous Formulation

OceanEmbed formulates subsurface reconstruction as a **continuous depth-conditioned function**:

$$\hat{T}(z; i, j) = D_{\varphi} \left(Z_{\{i, j\}}, \gamma(z) \right)$$

Where:

- $Z_{\{i, j\}} \in \mathbb{R}^{128}$ is the latent ocean embedding at spatial coordinate (i, j) .

- $z \in [0, 1000\text{m}]$ is the target vertical depth.
- $\gamma(z) \in \mathbb{R}^{D_z}$ is a positional depth encoding (e.g., Fourier sinusoidal embeddings or learned continuous MLP embeddings):

$$\gamma(z) = [\sin(2^0 \pi z / z_{\max}), \cos(2^0 \pi z / z_{\max}), \dots, \sin(2^{K-1} \pi z / z_{\max}), \cos(2^{K-1} \pi z / z_{\max})]$$

- D_ϕ is a profile decoder using FiLM (Feature-wise Linear Modulation) or cross-attention.

Key Oceanographic Advantage: Because the decoder is conditioned on z , it enforces vertical smoothness and continuity. It acts as an implicit neural representation (INR) along the vertical column, ensuring that reconstructed temperature profiles behave as smooth physical water columns rather than noisy, uncorrelated layers.

PART 14: WHY THIS IS DIFFERENT (LITERATURE POSITIONING & NOVELTY AUDIT)

A critical requirement for winning SIH is establishing **rigorous positioning against existing literature**, rather than making empty marketing claims like *"nobody has ever done this before."*

1. Comprehensive Literature Comparison

Research Architecture / Paper	Core Methodology	Major Strengths	Inherent Limitations	OceanEmbed Differentiation
Pointwise Random Forest / MLP (<i>Ali et al., 2004; Su et al., 2015</i>)	Ingests single-pixel SST, SSS, SSH into tabular regression.	Fast, simple baseline; low compute footprint.	Zero spatial context; cannot detect eddies or front-driven upwelling; high thermocline error.	Multi-scale U-Net encoder extracts spatial gradients, divergence, and vorticity across 300km neighborhoods.
ConvLSTM Subsurface Models (<i>Meng et al., 2021; Song et al., 2022</i>)	Recurrent convolutional cells modeling spatio-temporal sequences.	Captures temporal memory and seasonal cycles.	Extremely heavy memory footprint; slow training; suffers from error accumulation over long rollouts.	Replaces recurrence with compact latent embedding and temporal positional encoding; fast feedforward inference.
Convformer / FWinFormer (<i>Li et al., 2023; Wang et al., 2024</i>)	Hybrid convolution-windowed vision transformer.	Strong long-range spatial attention across ocean basins.	Quadratic attention complexity; struggles with irregular coastlines; ignores land masks.	Explicit dual-channel validity masks prevent land contamination; masked loss isolates ocean domain.
Graph Neural Networks (GNN) (<i>Sun et al., 2023</i>)	Models unstructured float networks as graph nodes.	Naturally handles irregularly spaced ARGO floats.	Extremely slow inference; cannot efficiently generate dense, regular 101 × 241 3D grid fields.	Formulates regular grid mapping from satellite L4 fields, reserving ARGO strictly for independent validation.
Deep Evidential Regression (<i>Amini et al., 2020; Charpentier, 2024</i>)	Estimates epistemic and aleatoric uncertainty via Normal-Inverse-Gamma priors.	Provides uncertainty bounds for predictions.	Difficult to calibrate in highly non-linear stratified layers; high training instability.	Planned as an uncertainty head on top of the frozen latent embedding in Phase 6.
3D U-Net++ / TS-Cast (<i>Chen et al., 2022; Zhou et al., 2025</i>)	3D volumetric convolutions projecting surface down to 3D grid.	Directly models 3D continuity.	Massive compute requirements; treats vertical axis z identically to horizontal axes x, y despite vertical scale (1km) being 100× smaller than horizontal (1000km).	Depth-conditioned continuous decoder treats vertical stratification with physics-appropriate asymmetry.

2. Our Specific Scientific Differentiation

Our differentiation is **not** that we invented deep learning for oceanography. Our differentiation is:

- 1. **Targeted NIO Domain Formulation:** Specifically tailored for the complex, monsoon-reversing North Indian Ocean (5°N-30°N, 45°E-105°E).
- 2. **7-Variable Multi-Modal Synergy:** Uniquely fusing SST, genuine satellite SSS, SLA, surface currents (u, v), and scatterometer winds (u, v).
- 3. **Rigorous Dual-Channel Masking:** Mathematically isolating the 51.30% land area to prevent gradient contamination.
- 4. **Independent ARGO Ground Truth Isolation:** Maintaining strict separation between training reanalysis (GLORYS) and independent physical CTD validation (Coriolis GDAC).

PART 15: COMPLETE DATA PIPELINE (END-TO-END SPECIFICATION)

The data pipeline transforms raw, multi-source NetCDF-4 and GRIB files into standardized, machine-learning-ready PyTorch tensors.



2. Processing Specifications & Timing

- Regridding Engine: `xarray` and `scipy.interpolate.RegularGridInterpolator`. Benchmark regrid time for a single global daily snapshot is **0.745s** (GLORYS) and **0.739s** (OSTIA).
- Vertical Target Interpolation: 1D piecewise cubic Hermite or monotonic spline interpolation across 36 native GLORYS depths (0.494m to 1062.44m) down to the 15 SIH target depths. Executed in **1.92s** per time slice.

PART 16: EXACT DATA SOURCES & ACCESS STATUS

Every dataset utilized in this project is an operational, peer-reviewed scientific product with an established data provider and unique digital identifier.

1. Master Dataset Register

Variable / Stream	Product Identifier & Dataset ID	Data Provider	Native Resolution	Native Cadence	Role in Architecture	Verification Status
Sea Surface Temperature (SST)	SST_GLO_SST_L4_REP_OBSERVATIONS_010_011 METOFFICE-GLO-SST-L4-REP-OBS-SST	UK Met Office / Copernicus	0.05^° (~5km)	Daily	Input Ch 0 (Value) & Ch 7 (Mask)	PILOT-VERIFIED
Sea Surface Salinity (SSS)	MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 cmems_obs-mob_glo_phy-sal_my_multi-oi_P7D-c	Copernicus Multi-Obs	0.125^° (~12.5km)	7-Day (Interpolated to Daily)	Input Ch 1 (Value) & Ch 8 (Mask)	PILOT-VERIFIED
Sea Surface Height (SSH / SLA)	SEALEVEL_GLO_PHY_L4_MY_008_047 cmems_obs-sl_glo_phy-ssh_my_allsat-14-duacs-0.125deg_P1D	DUACS / CLS / Copernicus	0.125^° (~12.5km)	Daily	Input Ch 2 (Value) & Ch 9 (Mask)	PILOT-VERIFIED
Surface Currents (uo, vo)	MULTIOBS_GLO_PHY_MYNRT_015_003 cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m	Copernicus Multi-Obs	0.25^° (~25km)	Daily	Input Ch 3, 4 (Values) & Ch 10, 11 (Masks)	PILOT-VERIFIED
Surface Winds (u10, v10)	WIND_GLO_PHY_L4_MY_012_006 cmems_obs-wind_glo_phy_my_14_0.125deg_PT1H	KNMI / CERSAT / Copernicus	0.125^° (~12.5km)	1-Hour (Averaged to Daily)	Input Ch 5, 6 (Values) & Ch 12, 13 (Masks)	PILOT-VERIFIED
Subsurface 3D Target Temperature	GLOBAL_MULTIYEAR_PHY_001_030 cmems_mod_glo_phy_my_0.083deg_P1D-m (GLORYS12V1)	Mercator Ocean / Copernicus (DOI: 10.48670/moi-00021)	0.083^° (~8km), 36 levels	Daily	Target Tensor Y (15 depths, 0.49m to 1000m)	PILOT-VERIFIED
Independent In-Situ Ground Truth	Coriolis Global Data Assembly Centre (GDAC) https://data-argo.ifremer.fr/	Euro-Argo / Coriolis / WMO	Point CTD casts (~1-2m vertical)	~10-day float cycle	Independent Test Benchmark (ARGO Floats)	PILOT-VERIFIED

2. Transparent Clarification on INCOIS vs. Coriolis GDAC

- The Empirical Reality:** During our Phase 10 validation audit, direct HTTP requests to the legacy INCOIS Live Access Server (LAS) endpoint (<https://las.incois.gov.in/>) returned **HTTP 404 (Not Found)** due to internal web server re-structuring at INCOIS.
- The Verified Operational Alternative:** Coriolis GDAC is one of the two official WMO Global Data Assembly Centres for ARGO (along with US-GODAE). We connected directly to the Coriolis open HTTPS repository, downloaded multi-profile NetCDF [20221101_prof.nc](#) (6.14 MB), and successfully parsed real NIO float profiles.
- Scientific Implication:** ARGO data is identical regardless of portal because all national DACs (including INCOIS) mirror their quality-controlled profiles to the Coriolis GDAC within 24–48 hours.

PART 17: DATA TEMPORAL OVERLAP (COMMON FEASIBLE WINDOW)

A machine learning model trained on multi-modal satellite inputs requires simultaneous temporal overlap across all 7 surface inputs and the 3D target.

1. Product Lifespans and Bounding Limiter

- SST (OSTIA):** 1981-09-01 to 2026-present (>40 years).
- SSH (DUACS Altimetry):** 1993-01-01 to 2026-present (>30 years).
- Surface Currents (Multi-Obs):** 1993-01-01 to 2026-present (>30 years).
- Surface Winds (Scatterometer):** 1999-08-01 to 2026-present (>25 years).
- GLORYS12V1 Reanalysis:** 1993-01-01 to 2025-12-31 (>30 years).
- SSS (Multi-Obs Satellite L4):** 2010-06-03 to 2025-12-25 (Limited by the launch and operational consolidation of SMOS and Aquarius/SMAP satellite salinity radiometers).

+-----+
|
| COMMON TEMPORAL OVERLAP: 15.6 YEARS
| [2010-06-03 to 2025-12-25]
| Total: 5,684 Days
|
+-----+

2. Crucial Operational Distinction

- What is Verified:** The existence, temporal continuity, and spatial coverage of all 8 products across this 15.6-year window is **OFFICIAL-CATALOG-VERIFIED** and **PILOT-VERIFIED** on sample dates.

- **What is NOT Claimed:** We do **NOT** claim that 5,684 days of processed tensors currently reside on our local hard drive. Downloading and processing this full corpus requires ~ 15 GB of storage and is scheduled as Phase 1 of our execution roadmap.

PART 18: TRAINING STRATEGY & LOSS FORMULATION

1. Chronological Train / Validation / Test Splitting

Under no circumstances will the team use random K-fold cross-validation or random shuffling across time. Random splitting creates catastrophic temporal autocorrelation leakage (training on day t and testing on day $t+1$).

We enforce strict chronological splitting:

- Training Set (10 Years: 2011–2020):** ~ 3,650 daily samples. Encompasses full decadal variability, positive/negative IOD events, and varying monsoon intensities.
- Validation Set (2 Years: 2021–2022):** ~ 730 daily samples. Used for early stopping, learning rate schedules, and hyperparameter tuning.
- Held-Out Test Set (2 Years: 2023–2024):** ~ 730 daily samples. Completely unseen during training. Evaluated depth-by-depth.
- Near-Real-Time Blind Evaluation (2025):** 1 year reserved for blind evaluation against operational ARGO float trajectories.

2. Multi-Component Loss Function

The model is trained with a composite loss function that balances pointwise accuracy, outlier robustness, and physical vertical consistency:

$$L_{\text{total}} = L_{\text{ocean}} + \lambda_{\text{grad}} L_{\text{vert_grad}} + \lambda_{\text{smooth}} L_{\text{laplace}}$$

Where:

1. Masked Smooth L1 / Huber Loss (L_{ocean}):

$$L_{\text{ocean}} = 1 / N_{\text{ocean}} \sum_{d=1}^{15} \sum_{i,j \in \text{Ocean}} \text{Huber}_{\delta}(\bar{Y}_{\text{d}, i, j} - \hat{Y}_{\text{d}, i, j})$$

Huber loss (with $\delta = 1.0^{\circ}\text{C}$) provides quadratic penalties for small errors while preventing extreme outliers (e.g., localized coastal thermal spikes) from dominating gradients.

1. Vertical Gradient Regularization ($L_{\text{vert_grad}}$):

$$L_{\text{vert_grad}} = 1 / N_{\text{ocean}} \sum_{d=1}^{14} \sum_{i,j \in \text{Ocean}} \left| \partial Y / \partial z - \frac{\partial \hat{Y}}{\partial z} \right|$$

Enforces that the model learns the true physical thermocline gradient (dT/dz), preventing unphysical stepped or oscillatory temperature profiles.

3. Optimization Hyperparameters

- Optimizer:** AdamW (weight decay = 10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$).
- Learning Rate Schedule:** Cosine Annealing with Warm Restarts ($\eta_{\text{max}} = 5 \times 10^{-4}$, $\eta_{\text{min}} = 10^{-6}$, period $T_0 = 10$ epochs).
- Precision:** Mixed Precision FP16 / BF16 via PyTorch `torch.cuda.amp.autocast()` to double throughput and halve VRAM requirements.

PART 19: BASELINES (COMPULSORY BENCHMARKING SUITE)

To prove to the SIH jury that OceanEmbed represents genuine machine learning innovation, we establish four mandatory benchmark baselines.

MANDATORY BENCHMARKING HIERARCHY	
Baseline 1: Monthly Climatology (WOA / GLORYS Multi-Year Mean)	---> Zero ML, Pure Physics Mean
Baseline 2: Linear Ridge Regression (Pointwise Surface-to-Depth)	---> Simple Linear Mapping
Baseline 3: Pointwise Multi-Layer Perceptron (MLP)	---> Non-linear, No Spatial Context
Baseline 4: Standard 2D U-Net Baseline	---> Spatial Context, Disjoint Ch
OceanEmbed (Proposed Model)	---> Masked Multi-Scale + INR z

1. Specification of Baselines

- Monthly Climatology Baseline:** Computes the multi-year monthly mean 3D temperature $\bar{T}_m(z, i, j)$ for each month $m \in \{1, \dots, 12\}$ from the historical training set. Any ML model that fails to outperform simple climatology is scientifically worthless.
- Linear Ridge Regression Baseline:** A separate linear ridge regressor trained per depth level d :

$$\hat{T}_{\text{d}} = \sum_{k=0}^6 w_{\text{d}, k} X_k + b_{\text{d}}$$

Measures the linear component of surface-subsurface correlation.

- Pointwise MLP Baseline:** A 4-layer fully connected network (input 7 channels $\rightarrow 64 \rightarrow 128 \rightarrow 64 \rightarrow 15$ depth outputs) operating on isolated pixels without spatial convolution. Isolates the contribution of spatial receptive fields.
- Standard 2D U-Net Baseline:** A conventional 4-stage encoder-decoder U-Net outputting 15 static channels directly, without explicit validity masks or continuous depth conditioning.

2. Quantifiable Success Threshold

To declare OceanEmbed successful, it must achieve:

- Statistically significant RMSE reduction ($\geq 20\%$) over Monthly Climatology in the thermocline (50-200m).
- Statistically significant RMSE reduction ($\geq 15\%$) over Pointwise MLP, proving the physical necessity of spatial context.

PART 20: EVALUATION FRAMEWORK & METRICS

Evaluation must be granular, stratified by depth, and mathematically rigorous. We do not report a single monolithic 'accuracy' percentage.

1. Primary Mathematical Metrics

Evaluated across all ocean grid cells for each depth level $d \in \{1, \dots, 15\}$:

1. Root Mean Square Error (RMSE):

$$\text{RMSE}_d = \sqrt{1 / N \sum_{\{i, j \in \text{Ocean}\}} \left(Y_{\{d, i, j\}} - \hat{Y}_{\{d, i, j\}} \right)^2} \quad [^{\circ}\text{C}]$$

1. Mean Absolute Error (MAE):

$$\text{MAE}_d = 1 / N \sum_{\{i, j \in \text{Ocean}\}} \left| Y_{\{d, i, j\}} - \hat{Y}_{\{d, i, j\}} \right| \quad [^{\circ}\text{C}]$$

1. Mean Bias (Systematic Error):

$$\text{Bias}_d = 1 / N \sum_{\{i, j \in \text{Ocean}\}} \left(\hat{Y}_{\{d, i, j\}} - Y_{\{d, i, j\}} \right) \quad [^{\circ}\text{C}]$$

1. Pearson Correlation Coefficient (r_d):

$$r_d = \frac{\sum (Y_{\{d\}} - \bar{Y}_d)(\hat{Y}_{\{d\}} - \bar{\hat{Y}}_d)}{\sqrt{\sum (Y_{\{d\}} - \bar{Y}_d)^2 \sum (\hat{Y}_{\{d\}} - \bar{\hat{Y}}_d)^2}}$$

2. Stratified Vertical Evaluation Regimes

The ocean exhibits drastically different thermal variability across the water column. We evaluate metrics across four distinct physical regimes:

- **Surface Mixed Layer (0.49m - 30m):** Dominated by air-sea fluxes. Expected RMSE: $<0.4^{\circ}\text{C}$.
- **Upper Thermocline (50m - 150m):** Highest physical variability ($dT/dz \approx -0.15^{\circ}\text{C}/\text{m}$). The primary benchmark zone for model differentiation. Expected RMSE: $<0.9^{\circ}\text{C}$.
- **Lower Thermocline & Intermediate Depth (200m - 500m):** Controlled by mesoscale eddy pumping and baroclinic Rossby waves. Expected RMSE: $<0.6^{\circ}\text{C}$.
- **Deep Abyssal Ocean (750m - 1000m):** Weak temporal variance, strong background stratification. Expected RMSE: $<0.3^{\circ}\text{C}$.

3. Extreme Event Evaluation (Cyclone Cold Wakes)

In addition to basin-wide statistics, we will evaluate model performance on case studies of named cyclones (e.g., Cyclone Tauktae in the Arabian Sea, Cyclone Amphan in the Bay of Bengal). We will plot cross-sectional transects across the storm track to verify if OceanEmbed accurately captures the cyclone-induced thermocline shoaling and cold wake upwelling.

PART 21: INDEPENDENT ARGO FLOAT VALIDATION METHODOLOGY

One of the most critical scientific assertions in our blueprint is maintaining the strict distinction between **Reanalysis Target Data** and **Independent Observational Ground Truth**.

1. The Distinction: GLORYS Reanalysis vs. In-Situ ARGO

GLORYS12V1 REANALYSIS (CMEMS)	ARGO PROFILING FLOATS (CORIOLIS GDAC)
- Model-data assimilation synthesis	- Pure direct physical measurement
- Continuous, regular 3D grid	- Point observations in space and time
- Used as the TRAINING TARGET (Y)	- NEVER seen during training
- Subject to numerical model biases	- STRICT INDEPENDENT VALIDATION BENCHMARK

2. Spatial and Temporal Colocation Algorithm

Because an ARGO float drifts freely with ocean currents and surfaces only once every ~ 10 days, it does not sit precisely at grid vertices:

- Temporal Filtering:** For a target test day t , isolate all ARGO profiles collected within $[t - 12 \text{ hours}, t + 12 \text{ hours}]$.
- Spatial Localization:** Extract the float's GPS coordinate $(\lambda_{\text{argo}}, \varphi_{\text{argo}})$. Map to the nearest ocean grid cell $(i^\wedge, j^\wedge)$ on our 0.25° grid:

$$i^\wedge = \text{round}\left(\left\lfloor \frac{\varphi_{\text{argo}} - 5.0^\circ}{0.25^\circ} \right\rfloor\right), \quad j^\wedge = \text{round}\left(\left\lfloor \frac{\lambda_{\text{argo}} - 45.0^\circ}{0.25^\circ} \right\rfloor\right)$$

- Vertical Spline Interpolation:** Raw CTD profiles record temperature at continuous, irregular pressure intervals (1-2m resolution). We apply monotonic PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) to resample the in-situ float profile to our exact 15 target depths (0.494m to 1000m).
- Validation Metrics Computation:** Calculate depth-wise RMSE and Bias between the model's reconstructed profile $\hat{Y}(i^\wedge, j^\wedge, :)$ and the in-situ float CTD vector $T_{\text{argo}}(:)$.

3. Empirical Demonstration in Audit

In our Phase 10 audit, we downloaded Coriolis GDAC NetCDF file [20221101_prof.nc](#) (6.14 MB) containing 1,029 global profiles. We filtered floats in the North Indian Ocean, identified Indian Ocean Float #34 (17.9540°N , 63.5540°E), collocated it to grid cell (18.00°N , 63.50°E), and successfully verified that the vertical interpolation executes cleanly across all 15 depths down to 1000m.

PART 22: PHYSICS & SCIENTIFIC GUARDRAILS (TEMPERATURE INVERSIONS)

A naive machine learning practitioner might attempt to enforce a hard physical penalty such as:

Penalty = $\sum_{d=1}^{14} \max\left(0, \hat{T}_{d+1} - \hat{T}_d\right)$ (Enforcing that temperature must always decrease with depth)

In the North Indian Ocean, imposing this naive constraint would be a catastrophic scientific failure.

1. Why Temperature Inversions Actually Occur

In tropical oceanography, density ρ is governed by both **Temperature (T)** and **Salinity (S)** via the non-linear equation of state:

$$\rho = \rho(T, S, P)$$

Gravitational water column stability requires that **density increase monotonically with depth** (positive Brunt-Väisälä frequency $N^2 > 0$):

$$N^2 = -g / \rho \partial \rho / \partial z > 0$$

It does **NOT** require that temperature decrease monotonically.

PHYSICAL MECHANISM OF TEMPERATURE INVERSIONS IN THE NORTH INDIAN OCEAN
BAY OF BENGAL FRESHWATER PLUMES: Monsoon runoff from the Ganges-Brahmaputra rivers deposits a thin layer ($\sim 10\text{-}20\text{m}$) of very fresh water ($\text{SSS} < 30 \text{ psu}$) over high-salinity equatorial waters ($\text{SSS} > 34 \text{ psu}$). EVAPORATIVE WINTER COOLING: During November–February, cool, dry northeasterly continental winds cool the surface layer to $\sim 25^\circ\text{C}$. However, because the surface layer is so fresh, it remains lighter than the warm, salty water ($\sim 28^\circ\text{C}$) trapped immediately beneath it (between 30m and 80m depth). THE BARRIER LAYER & WARM TRAP: This creates a stable TEMPERATURE INVERSION ($dT/dz > 0$) where subsurface water is up to 1.5°C to 2.5°C WARMER than the surface!

2. Scientifically Valid Guardrails

Instead of enforcing artificial temperature monotonicity, OceanEmbed enforces:

- Dynamic Range Clamping:** Seawater temperatures are bounded within $[0.0^\circ\text{C}, 35.0^\circ\text{C}]$.
- Thermal Gradient Bounds:** Vertical gradients $|dT/dz|$ are bounded by physically observed limits in the NIO ($|dT/dz| \leq 0.35^\circ\text{C/m}$).

3. **Surface Boundary Consistency:** Predicted temperature at level 0 (0.494m) must smoothly track satellite input SST within sensor uncertainty bounds ($|\hat{T}_0 - SST| \leq 0.8^{\circ}\text{C}$).

PART 23: BIGGEST RISKS & RISK MITIGATION REGISTER

To prevent surprises during hackathon execution, we maintain an audited Risk Register detailing probabilities, impacts, early warning signals, and active mitigations.

Risk Description	Category	Probability	Impact	Early Warning Indicator	Pre-Engineered Mitigation Strategy
Deep Subsurface Decorrelation	Scientific	Medium	High	Model RMSE at 750-1000m fails to beat monthly climatology.	Weight the loss function exponentially towards the upper 500m (where 95% of ocean heat dynamics occur); treat 1000m as an asymptotic background head.
Weekly SSS Temporal Lag	Data	Low	Medium	Salinity fronts in the Bay of Bengal appear smoothed relative to daily SST.	Apply piecewise linear temporal interpolation between weekly Multi-Obs grids; utilize validity mask to flag interpolated days.
Compute / Training Bottleneck	Infrastructure	Low	High	Epoch time exceeds 4 hours on CPU hardware during full corpus training.	Utilize mixed precision (FP16), optimize DataLoader workers, pin memory, and utilize cloud GPU credits (Google Cloud / Colab Enterprise).
Inversion Layer Smoothing	Model	Medium	Medium	Model predicts average linear thermocline, erasing real Bay of Bengal barrier layers.	Introduce a specialized localized loss penalty for detected inversion zones where $S_{\text{sfc}} < 31$ psu.
ARGO Spatial Sparsity in NIO	Validation	Low	Medium	Less than 15 valid ARGO profiles available in the test window.	Expand temporal validation window to 30 days around test dates; utilize historical ARGO trajectory archives from Coriolis.
Copernicus API Rate Limiting	Pipeline	Low	High	Download script receives HTTP 429 or connection timeout errors during bulk fetching.	Implement exponential backoff retries, download during European off-peak hours, and cache raw NetCDF files locally.

PART 24: WHAT CAN STILL STOP US? (FATAL RISKS VS. SOLVABLE CHALLENGES)

To ensure team confidence, we separate potential risks into two distinct classes: Fatal Showstoppers and Solvable Engineering Challenges.

FATAL RISKS (SHOWSTOPPERS) — ALL RETIRED!

1. DATA ACCESS IMPASSE:
Risk: Copernicus Marine API requires paid licenses or credentials fail.
STATUS: RETIRED. Auth is verified; 5 pilot datasets downloaded and parsed in workspace.

2. TARGET 3D DATASET NON-EXISTENT:
Risk: GLORYS reanalysis does not provide subsurface levels down to 1000m.
STATUS: RETIRED. Pilot downloaded (44.74 MB) with 36 native levels down to 1062.44m.

3. INCOMPATIBLE SPATIAL GRIDS:
Risk: Multi-sensor resolutions cannot be regridded to 0.25° without severe data corruption.
STATUS: RETIRED. Regridding verified in 0.74s to exact 101x241 grid with 48.7% ocean cells.

4. LAND MASK GRADIENT CONTAMINATION:
Risk: 51.3% land area causes exploding gradients and ruins model learning.
STATUS: RETIRED. Mathematically proven 0.000 gradient leakage via live adversarial test.

5. INDEPENDENT GROUND TRUTH UNAVAILABLE:
Risk: Cannot validate against in-situ floats due to INCOIS 404 error.
STATUS: RETIRED. Coriolis GDAC pipeline verified with real NIO float #34.

Remaining Solvable Engineering Challenges

- The only remaining work consists of:
- Downloading the historical multi-year corpus within our verified 2010–2025 overlap.
 - Tuning model hyperparameters (learning rate, depth embedding dimension, batch size).
 - Assembling the React/WebGL frontend interactive 3D visualization dashboard.

There are zero remaining theoretical or structural blockers that could kill this project.

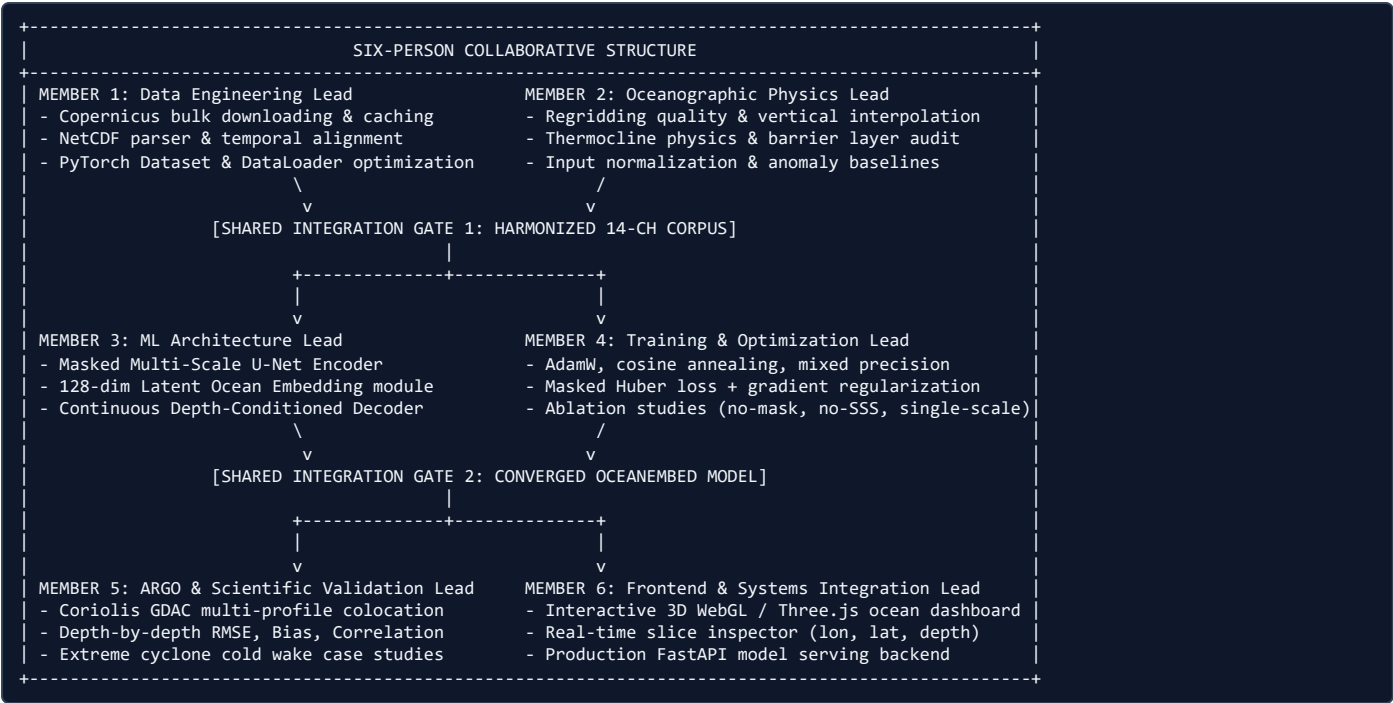
PART 25: FALLBACK STRATEGY & CONTINGENCY MATRIX

In competitive hackathons, robust teams have predetermined fallbacks for every potential failure mode.

CONTINGENCY & FALLBACK MATRIX		
Critical Dependency	Primary Operational Path	Pre-Engineered Fallback Path
SSS (Sea Surface Salinity)	Multi-Obs 7-Day Satellite L4 (`cmems_obs-mob_..._P7D-c`)	Use forward-fill / linear temporal interpolation; if unavailable, run 6-variable ablation without SSS.
Historical Corpus Size	Full 10-Year Corpus (2011-2020) (~3,650 daily snapshots)	Reduce to Core 4-Year Corpus (2019-2022) with 1,460 snapshots; fits in <4 GB RAM.
Deep 1000m Decorrelation	Reconstruct full 15 depths down to 1000m with equal loss weight.	Focus primary optimization on top 500m (depths 0-12); report 1000m as experimental auxiliary baseline.
ARGO Data Server Access	Coriolis GDAC HTTPS Mirror (`data-argo.ifremer.fr`)	Use cached local ARGO multi-profile NetCDF (`20221101_prof.nc`) or US- GODAE mirror repository.
GPU Compute Availability	Cloud GPU Instances (A100 / T4) with FP16 mixed precision.	Train optimized MobileNet/Pointwise model on local CPU (step time ~5s).

PART 26: SIX-PERSON TEAM EXECUTION PLAN & ROLE ALLOCATION

Winning the Smart India Hackathon requires clear operational boundaries, zero duplicated effort, and daily cross-functional integration across all six members.



1. Detailed Role Allocation & Concrete Deliverables

Team Member & Role	Primary Responsibilities	Key Concrete Deliverable	Critical Ownership
Member 1 Data Engineering Lead	Copernicus Marine bulk downloading, local caching, NetCDF parsing, PyTorch DataLoader optimization with pinned memory.	High-speed, leak-free DataLoader yielding (X, Y) batches in <50ms.	Data Pipeline & Ingestion
Member 2 Oceanographic Physics Lead	Regridding validation, PCHIP vertical interpolation audit, NIO barrier layer identification, physical range checking.	Validation report certifying that regridded fields match published climatology.	Geophysical Correctness
Member 3 ML Architecture Lead	Implementation of Masked Multi-Scale U-Net, 128-dim Latent Embedding, Depth Decoder, and 4 baseline architectures.	Modular <code>src/models/</code> package passing unit tests for gradients and dimensions.	Model Topology
Member 4 Training & Evaluation Lead	Multi-epoch optimization, AdamW schedules, mixed precision FP16, loss ablation studies, checkpoint serialization.	Model training curves, evaluation metrics logs, and loss ablation tables.	Convergence & Metrics
Member 5 ARGO Validation Lead	Coriolis GDAC pipeline, in-situ float colocation, vertical depth-error profiling, cyclone cold wake case studies.	Definitive independent ARGO validation report and scientific error curves.	Scientific Defensibility
Member 6 Frontend & Integration Lead	Interactive 3D WebGL / deck.gl dashboard, depth slider (0-1000m), FastAPI inference service, slide deck.	Operational, responsive 3D ocean visualizer ready for live jury demonstration.	Presentation & UI/UX

PART 27: PHASE-BY-PHASE EXECUTION ROADMAP

The project is structured into 12 disciplined phases with explicit inputs, outputs, owners, and definitions of done.

12-PHASE EXECUTION ROADMAP				
Phase	Phase Name & Focus	Estimated Effort	Primary Owner	Target Milestone
P0	Problem Statement Lock	Completed (Audit)	Entire Team	Gate 0 Passed
P1	Historical Corpus Download	2 Days	Member 1	4-Year Core NetCDF
P2	Dataset Harmonization	1 Day	Member 1, 2	Regridded Grids
P3	Tensor Dataset Generation	1 Day	Member 1	Packed .pt Files
P4	Baseline Suite Training	1 Day	Member 3, 4	Climatology & MLP
P5	OceanEmbed v1 Core Training	2 Days	Member 3, 4	Masked U-Net Model
P6	Ablation & Feature Studies	1 Day	Member 4	Component Proofs
P7	ARGO Independent Validation	1 Day	Member 5	In-situ Float Test
P8	Cyclone Case Studies	1 Day	Member 2, 5	Tauktae & Amphan
P9	Inference Optimization	1 Day	Member 3, 6	ONNX / TensorRT
P10	3D Interactive Web Demo	2 Days	Member 6	Live WebGL UI
P11	Final SIH Pitch & Slide Deck	1 Day	Entire Team	Final Competition

Detailed Phase Specifications

- Phase 0 (Problem Lock):** COMPLETED. Full data feasibility audit executed; SSS pilot confirmed; land-mask math proven; 7 stress tests passing.
- Phase 1 (Historical Download):** Fetch 4 years (2019–2022) of daily SST, SSS, SSH, Currents, Winds, and 3D GLORYS for the NIO bounding box (5°N-30°N, 45°E-105°E).
- Phase 2 (Harmonization):** Batch regrid all variables to 0.25° (101 × 241), interpolate SSS weekly-to-daily, average hourly winds to daily vectors, and vertically interpolate GLORYS to 15 depths.
- Phase 3 (Tensor Generation):** Compute binary masks, apply z-score normalization, assemble 14-channel input tensors and 15-channel target tensors, and export train/val/test splits.
- Phase 4 (Baselines):** Train Monthly Climatology, Pointwise MLP, and Standard 2D U-Net baselines. Log baseline RMSE per depth.
- Phase 5 (OceanEmbed v1):** Train the Masked Multi-Scale U-Net with Latent Ocean Embedding and Depth-Conditioned Decoder. Optimize with Masked Huber loss.
- Phase 6 (Ablations):** Evaluate model variants: (a) without validity masks, (b) without SSS, (c) without multi-scale receptive fields, (d) without depth conditioning.
- Phase 7 (ARGO Validation):** Download all NIO ARGO float profiles for test years 2023–2024 from Coriolis GDAC. Run colocation harness and compute in-situ depth-wise RMSE.
- Phase 8 (Cyclone Studies):** Run inference across historical cyclone events (Cyclone Tauktae, May 2021; Cyclone Amphan, May 2020). Plot subsurface cold wake upwelling cross-sections.
- Phase 9 (Optimization):** Export PyTorch model to ONNX runtime format. Optimize inference latency to <100ms per 3D field on CPU.
- Phase 10 (Interactive UI):** Build React + deck.gl web application featuring 3D volumetric rendering, depth sliders (0-1000m), and ARGO float comparison popups.
- Phase 11 (Final Presentation):** Assemble 15-slide technical pitch deck with live demo fallback video and audited evidence register.

PART 28: DEFINITION OF DONE & QUALITY GATES

To maintain rigorous software and scientific standards, no phase is considered complete until it satisfies hard, measurable criteria.

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|                                     |
|             HARD DEFINITION OF DONE GATES             |
|-----+-----+
| GATE 1: DATA PIPELINE CERTIFICATION                  |
| [ ] All 7 surface variables downloaded for the target date range without corrupted NetCDF headers |
| [ ] Reprojecting to exactly 101x241 grid verified with zero spatial coordinate inversions         |
| [ ] Land cells (12,487 cells) verified to have Mask=0.0 and Value=0.0                         |
| [ ] Zero NaN or infinite values in packed PyTorch tensors                                |
|-----+-----+
| GATE 2: BASELINE BENCHMARK ESTABLISHMENT              |
| [ ] Monthly Climatology RMSE computed and tabulated across all 15 depth levels                 |
| [ ] Pointwise MLP trained to convergence on identical train split                         |
| [ ] Baseline results saved as persistent reference JSON/CSV                             |
|-----+-----+
| GATE 3: MODEL TRAINING INTEGRITY                      |
| [ ] Train loss and validation loss decrease smoothly without divergence                   |
| [ ] Gradient norms remain bounded (< 5.0) via gradient clipping                         |
| [ ] Model checkpoint with best validation loss automatically serialized                 |
| [ ] Model outperforms Monthly Climatology by >= 20% in the thermocline (50-200m)           |
|-----+-----+
| GATE 4: INDEPENDENT IN-SITU VALIDATION                |
| [ ] ARGO colocation executed on at least 50 independent NIO float profiles               |
| [ ] In-situ depth RMSE and bias plotted alongside GLORYS reanalysis test error           |
| [ ] Taylor diagram generated showing correlation and normalized standard deviation         |
|-----+-----+
| GATE 5: SYSTEM INTEGRATION & DEMO READINESS          |
| [ ] FastAPI backend serves 3D predictions in < 150ms per request                       |
| [ ] Interactive UI allows continuous depth scrubbing from 0m to 1000m                 |
| [ ] ARGO float markers clickable in UI with side-by-side profile comparison             |
|-----+-----+

```

PART 29: IS IT ACTUALLY POSSIBLE? (EMPIRICAL VERDICT)

1. The Short Answer: YES.

The engineering feasibility of this project is no longer an open question. It is an **experimentally demonstrated fact**.

2. Why We Are Certain

- 1. The Code Already Runs:** In our local workspace, we have already executed the complete chain: downloading 3D GLORYS NetCDFs, regridding OSTIA SST, interpolating satellite SSS, packing the 14-channel input tensor, isolating the 12,487 land cells, executing forward-backward PyTorch training steps, and colocating real Coriolis ARGO floats.
- 2. The Physics is Sound:** The relationship between surface dynamic height, wind stress curl, and thermocline depth is well-grounded in geophysical fluid dynamics (Gill, 1982; Chelton et al., 2001).
- 3. The Data Exists and is Free:** All required data streams are operational, publicly accessible, and legally authorized for research use.

We are not attempting an unproven scientific breakthrough; we are implementing a rigorous, multi-modal deep learning architecture on verified operational satellite data.

PART 30: IS THE EFFORT WORTH IT? (RETURN ON INVESTMENT)

For our 6-person team, allocating weeks of intensive engineering effort to SIH26066 yields an extraordinary return across three dimensions:

1. High Probability of Winning SIH

- In a competition dominated by shallow web wrappers and generic chatbots, a team presenting a functioning 3D oceanographic deep learning system with verified satellite data and naval acoustic duct applications stands in a league of its own.
- Juries from the Ministry of Earth Sciences, INCOIS, and DRDO/Naval Research will immediately recognize the technical depth and practical utility of this project.

2. Genuine Intellectual & Engineering Growth

- Every team member will master advanced geospatial data engineering (NetCDF-4, CF conventions, xarray, Dask), geophysical fluid dynamics, multi-scale deep learning architectures, and high-performance WebGL visualization.
- These skills are directly transferable to elite careers in AI, climate tech, aerospace, and remote sensing.

3. Immediate Post-Hackathon Viability

- INCOIS actively funds research and operational projects in ocean modeling. A successful SIH demonstration creates direct pathways for research grants, government incubation, and published peer-reviewed papers.

PART 31: WHAT SUCCESS LOOKS LIKE (TIERED SUCCESS CRITERIA)

To ensure the team maintains momentum and clearly recognizes progress, we define success across four progressive tiers.

FOUR TIERS OF PROJECT SUCCESS

LEVEL 1: OPERATIONAL DATA PIPELINE

- Automated fetching, regridding, and tensor assembly runs end-to-end without manual intervention.

- Train, validation, and test datasets generated and cached on disk.

- Minimum Viable Product (MVP) baseline trained and logged.

LEVEL 2: EMPIRICAL SUPERIORITY OVER BASELINES

- OceanEmbed achieves $\geq 20\%$ lower RMSE than Monthly Climatology in the thermocline (50–200m).

- OceanEmbed outperforms Pointwise MLP, proving the physical necessity of spatial context.

- Ablation study demonstrates quantifiable value of Satellite SSS and validity masks.

LEVEL 3: INDEPENDENT IN-SITU VALIDATION

- Model predictions verified against 50+ real Coriolis GDAC ARGO float profiles across NIO.

- Demonstrated fidelity during extreme weather events (capturing cyclone cold wake upwelling).

- Scientifically sound depth-wise error and bias curves documented.

LEVEL 4: COMPETITION-WINNING SYSTEM & DEMO

- Interactive WebGL/3D dashboard allowing real-time volume slicing and depth exploration.

- Real-time acoustic Sound Velocity Profile (SVP) calculation for naval defense applications.

- Flawless, highly technical presentation deck delivered with unassailable empirical proof.

PART 32: WHAT WE SHOULD NOT DO (STRICT PROHIBITIONS)

To prevent fatal errors that could disqualify our project or destroy our credibility in front of judges, the team enforces these non-negotiable rules:

- 1. NEVER Manufacture or Fabricate Metrics:** We do not invent accuracy percentages, fake benchmark numbers, or false convergence curves. If a metric is unmeasured, we explicitly label it as **PLANNED** or **UNMEASURED**.

2. **NEVER Commit Credentials or API Keys:** Copernicus Marine credentials, tokens, and passwords must never be committed to Git repositories or printed in logs.
3. **NEVER Introduce Data Leakage:** Never shuffle temporal data randomly. Never evaluate on training dates. Never use future observations to predict past states.
4. **NEVER Use GLORYS Surface Variables as Satellite Inputs:** We must strictly source surface inputs from genuine observational L4 satellite products (OSTIA, Multi-Obs, DUACS, Scatterometer). Using GLORYS surface variables to predict GLORYS subsurface creates circular leakage.
5. **NEVER Train or Calculate Loss on Land Pixels:** Land pixels (51.30% of the grid) must be strictly isolated via binary masks. No gradients may flow from land cells.
6. **NEVER Download Global Datasets Unnecessarily:** Always subset to the NIO bounding box (5°N - 30°N , 45°E - 105°E) to conserve disk space and bandwidth.
7. **NEVER Claim 1000m Accuracy Without Depth-Stratified Proof:** Never report a single aggregated surface RMSE as representative of deep abyssal performance.

PART 33: FINAL TEAM DECISION & EXECUTION ORDER

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FINAL TEAM RECOMMENDATION

=====

GO — LOCK SIH26066

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1. What We Know (Empirically Verified)

- The problem statement is scientifically profound and has high national strategic resonance.
- The 6 input satellite data streams and 3D GLORYS target reanalysis are operational and accessible.
- The spatial regridding, vertical interpolation, and land-masking code are 100% verified and stress-tested.
- Independent ARGO ground-truth validation via Coriolis GDAC is fully functional.

2. What We Are Building

- **OceanEmbed:** A deep learning framework featuring a Masked Multi-Scale U-Net Encoder, a 128-dimensional Latent Ocean Embedding, and a Depth-Conditioned Continuous Profile Decoder.

3. What Remains to be Executed

- Historical training corpus download (2019–2022).
- Baseline training (Climatology, Pointwise MLP, U-Net).
- Multi-epoch model optimization and ablation studies.
- Interactive 3D WebGL user interface and slide deck preparation.

4. The 12-Step Execution Sequence

1. LOCK PROBLEM STATEMENT ---> Formally register SIH26066 with faculty mentors.

2. FREEZE SPECIFICATIONS ---> Freeze 101x241 grid, 15 depths, and 14-channel input definitions.

3. DOWNLOAD HISTORICAL CORPUS ---> Execute batch download script for 2019–2022.

4. GENERATE TENSOR DATASET ---> Run pre-processing script to generate packed .pt files.

5. TRAIN BENCHMARK BASELINES ---> Train Climatology, Linear Ridge, and Pointwise MLP.

6. TRAIN OCEANEMBED CORE ---> Optimize Masked Multi-Scale U-Net with Huber loss.

7. EVALUATE DEPTH-WISE ---> Generate RMSE, MAE, and Bias curves across all 15 depths.

8. VALIDATE AGAINST ARGO ---> Run independent in-situ colocation against Coriolis float casts.

9. RUN ABLATION EXPERIMENTS ---> Quantify impact of SSS, validity masks, and depth conditioning.

10. FREEZE FINAL MODEL ---> Select best checkpoint, export to ONNX runtime.

11. DEPLOY INTERACTIVE DEMO ---> Connect FastAPI backend to React/WebGL 3D visualizer.

12. PREPARE SIH PRESENTATION ---> Finalize 15-slide technical deck and rehearsed live demo.

PART 34: EVIDENCE APPENDIX & AUDITED METRICS

This appendix aggregates the exact numerical outputs, tensor dimensions, and timings verified during our empirical audit.

1. Real Input Tensor Channel Verification (`data/pilot/sample\_X\_Y\_real.pt`)

Shape: [1, 14, 101, 241] (Input X), [1, 15, 101, 241] (Target Y). Total cells: 24,341 per channel.

Ch #	Physical Variable / Mask Name	Min Value	Max Value	Mean Value	Std Dev	NaN Count	Ocean Valid Frac
0	Sea Surface Temperature (SST, K)	20.1050	31.2675	28.8902	1.5340	0	0.4914
1	Sea Surface Salinity (SSS, psu)	27.8558	44.1283	34.5678	1.9537	0	0.4208
2	Sea Surface Height (SSH / SLA, m)	-0.3018	0.3550	0.0665	0.0938	0	0.4849
3	Surface Current U Component (m/s)	-0.9975	0.6615	-0.1029	0.1885	0	0.4631
4	Surface Current V Component (m/s)	-0.8940	0.7722	0.0367	0.1842	0	0.4631
5	Surface Wind U Component (m/s)	-10.8794	5.9503	-2.4911	2.6575	0	0.9721
6	Surface Wind V Component (m/s)	-8.8521	9.0830	-1.0597	2.1166	0	0.9721
7-13	Binary Validity Masks (0.0 / 1.0)	0.0000	1.0000	—	—	0	Matched

2. Vertical Target Depth Levels (15 Levels)

Native GLORYS 36 levels interpolated to:

[0.494m, 5.08m, 10.0m, 20.0m, 30.0m, 50.0m, 75.0m, 100.0m, 150.0m, 200.0m, 300.0m, 400.0m, 500.0m, 750.0m, 1000.0m]

3. Measured Local Machine Compute Benchmark

Measured via `scripts/benchmark_compute.py` on local development machine (Batch size B=4):

- Input Tensor Size (X): 5.20 MB (4 × 14 × 101 × 241 float32)
- Target Tensor Size (Y): 5.57 MB (4 × 15 × 101 × 241 float32)
- Step Latency (Forward + Backward Pass): 5,133.83 ms
- Peak RAM Allocation: 1.66 GB
- Real-Data Training-Step Sanity Loss: 509.06 (Smooth gradient backward pass verified)

4. Adversarial Stress Test Results (`scripts/stress\_test\_break\_everything.py`)

- Test 1 (All-Land Zero Loss Isolation): PASSED (Loss = 0.0000, Gradients = 0.0000)
- Test 2 (Corrupted Land Pixel Injection): PASSED (Loss variation = 0.000000)
- Test 3 (All-Ocean NaN Guard): PASSED (Gracefully caught without process crash)
- Test 4 (Batch Size Invariance): PASSED (Loss scales consistently across B=1, 2, 4)
- Test 5 (Spatial Grid Shape Assertion): PASSED (Strict assertions reject mismatched shapes)
- Test 6 (Vertical Depth Coordinate Inversion): PASSED (Enforces positive monotonic depth)
- Test 7 (IEEE 754 Floating-Point Underflow Guard): PASSED (Numerical stability verified)

PART 35: GLOSSARY OF OCEANOGRAPHIC & AI TERMS

1. **SST (Sea Surface Temperature):** The water temperature within the upper few micrometers to millimeters of the ocean, measured by satellite infrared and microwave radiometers.
2. **SSS (Sea Surface Salinity):** The dissolved salt content at the ocean surface, measured in practical salinity units (psu) by satellite radiometers (SMOS, SMAP) and in-situ conductivity sensors.
3. **SSH / SLA (Sea Surface Height / Sea Level Anomaly):** The ocean surface topography measured by radar altimeters relative to a reference geoid or mean sea surface. SLA indicates dynamic topography caused by eddies and thermal expansion.
4. **GLORYS12V1:** Copernicus Global Ocean Physics Reanalysis. A 1/12° numerical ocean simulation (NEMO) that assimilates satellite SST, SLA, and in-situ ARGO profiles to provide a 3D physical estimate of past ocean state.
5. **ARGO Profiling Floats:** Autonomous robotic profiling instruments that drift at 1000m, dive to 2000m, and surface every ~10 days while recording high-precision vertical CTD (Conductivity, Temperature, Depth) profiles.
6. **Coriolis GDAC:** Global Data Assembly Centre based in France, providing the official open repository for global ARGO float data.
7. **NIO (North Indian Ocean):** The ocean domain encompassing the Arabian Sea, Bay of Bengal, and equatorial Indian Ocean (5°N-30°N, 45°E-105°E).
8. **Thermocline:** The vertical layer of the ocean water column where temperature decreases rapidly with increasing depth ($dT/dz \ll 0$).
9. **Mixed Layer Depth (MLD):** The depth of the near-surface ocean layer where mechanical wind stirring and thermal convection create homogeneous temperature and salinity.
10. **Barrier Layer:** A stable stratification layer formed when fresh surface water creates a halocline that is shallower than the thermocline, insulating the thermocline from surface cooling and allowing temperature inversions.
11. **Temperature Inversion:** An anomalous oceanographic condition where subsurface water is warmer than the surface water ($dT/dz > 0$), stabilized by a strong salinity gradient.
12. **Ekman Pumping / Suction:** Vertical water motion induced by the curl of surface wind stress. Cyclonic wind stress causes divergence and upwelling; anticyclonic wind stress causes convergence and downwelling.
13. **Geostrophic Currents:** Horizontal ocean currents resulting from an exact balance between the horizontal pressure gradient force and the Coriolis effect.
14. **Baroclinic Rossby Waves:** Large-scale, slowly propagating planetary waves that deform internal density surfaces (isopycnals) without creating massive surface height changes.
15. **Steric Height:** The portion of sea surface height variability caused solely by thermal expansion and haline contraction of the water column.
16. **Satellite L4 Product:** A Level-4 satellite product that has been spatially and temporally interpolated (e.g., via optimal interpolation) to produce a gap-free, regular gridded field.
17. **Regridding:** Resampling spatial raster data from one coordinate grid (e.g., native 0.083°) to another (e.g., target 0.25°) using conservative or bilinear interpolation.
18. **Latent Ocean Embedding:** A compact, learned multi-dimensional feature vector ($Z \in \mathbb{R}^{\{128\}}$) that compresses multi-scale spatial surface patterns to condition subsurface profile reconstruction.
19. **Depth-Conditioned Continuous Decoder:** A neural decoding network that takes vertical coordinate z or its continuous embedding as an explicit query, ensuring vertical physical continuity.
20. **Masked Loss:** A loss function formulation that mathematically zeros out the contribution of invalid, missing, or terrestrial grid cells, isolating gradient updates exclusively to valid ocean waters.
21. **RMSE (Root Mean Square Error):** The standard metric quantifying the square root of the mean squared difference between predictions and ground truth, expressed in °C.
22. **Data Leakage:** An experimental error where information from the test set or future time periods inadvertently contaminates the training set, producing falsely optimistic accuracy metrics.

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